

Benchmarking LLM-Generated Biological Hypotheses for Scientific Discovery



Project
Website

Jefferson Chen
jec068@ucsd.edu

Samuel Lee
hsl023@ucsd.edu

Jieyuan Liu
jil029@ucsd.edu

Zhiting Hu
zh019@ucsd.edu

Zhen Wang
zhw085@ucsd.edu

UC San Diego™
HALICIOĞLU DATA SCIENCE INSTITUTE

Introduction

- **Problem:** Single-pass LLM prompting produces **one** hypothesis with no systematic refinement.
- **Gap:** Unlike researchers, LLMs lack **selection pressure** to keep strong ideas and revise weak ones.
- **Idea:** Treat hypothesis refinement as **evolutionary optimization** (selection, crossover, mutation) [1].
- **HypoEvolve:** Genetic Algorithm (GA) + multi-agent framework
- **Drug Repurposing (DR)** with DepMap CRISPR dependency
- **Type 2 Diabetes (T2D)** associated gene identification

Methods

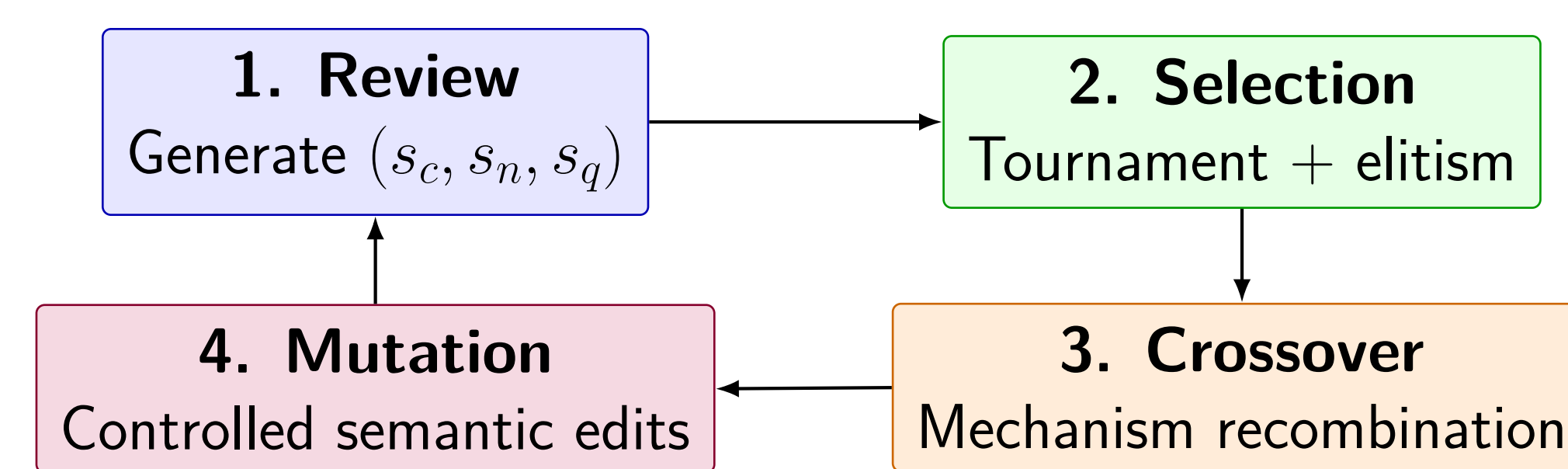


Figure 1: Genetic algorithm cycle per generation.

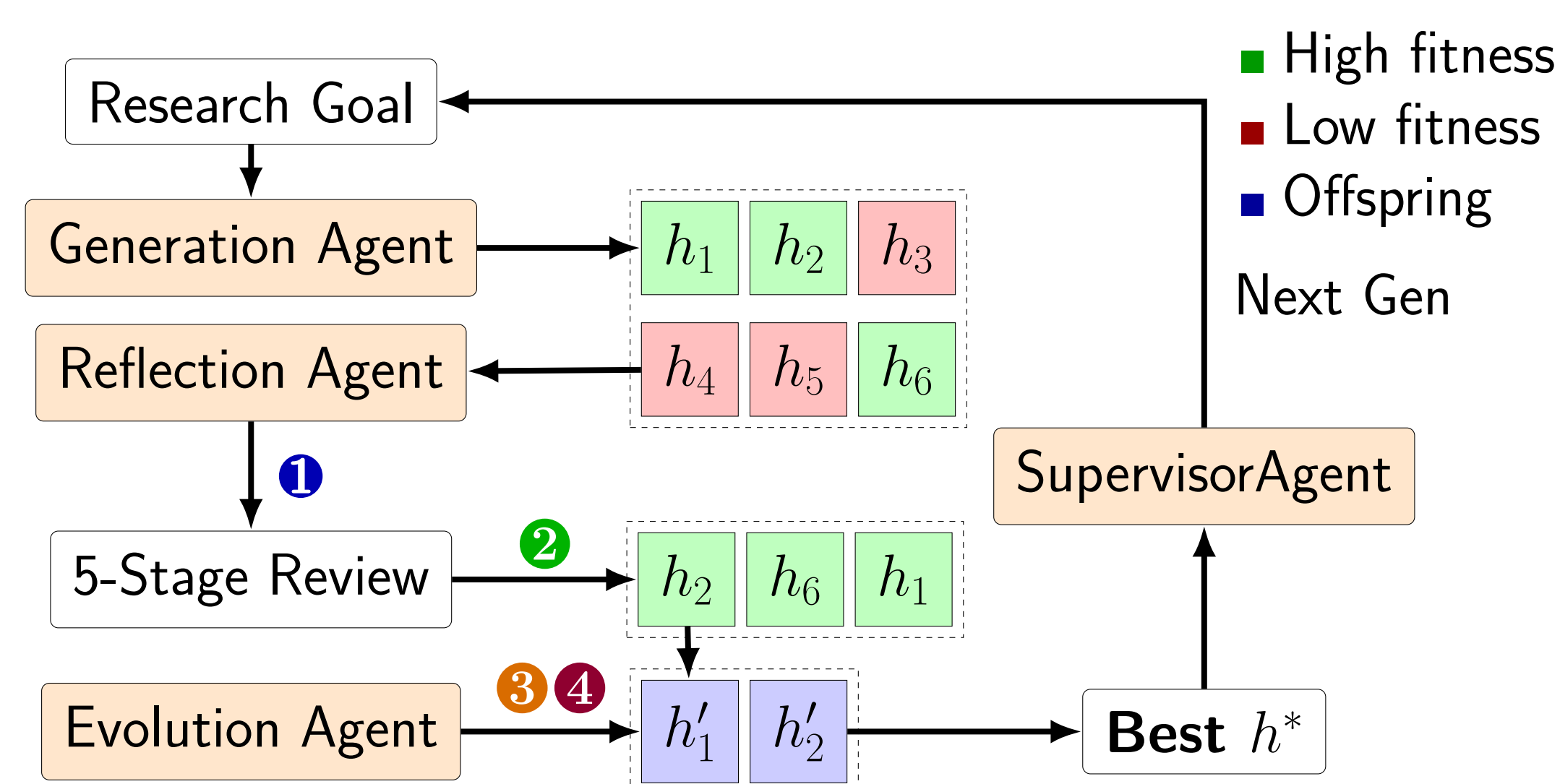


Figure 2: HypoEvolve multi-agent evolutionary loop.

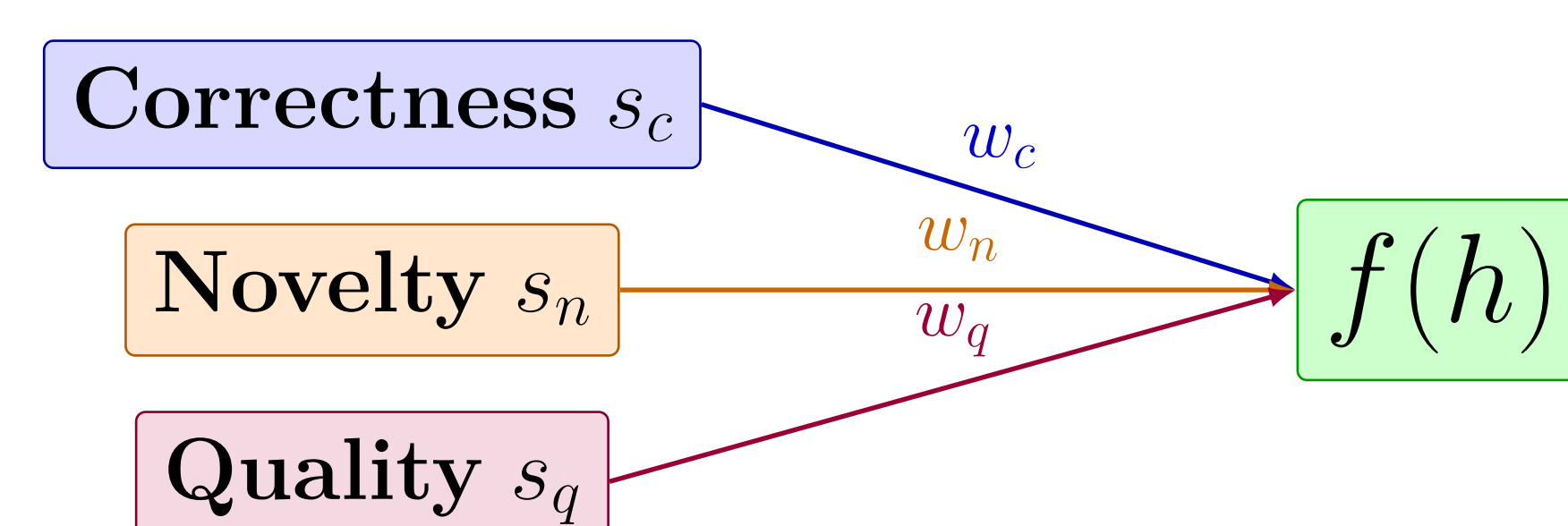


Figure 3: Fitness aggregation from LLM-evaluated component scores.

Results

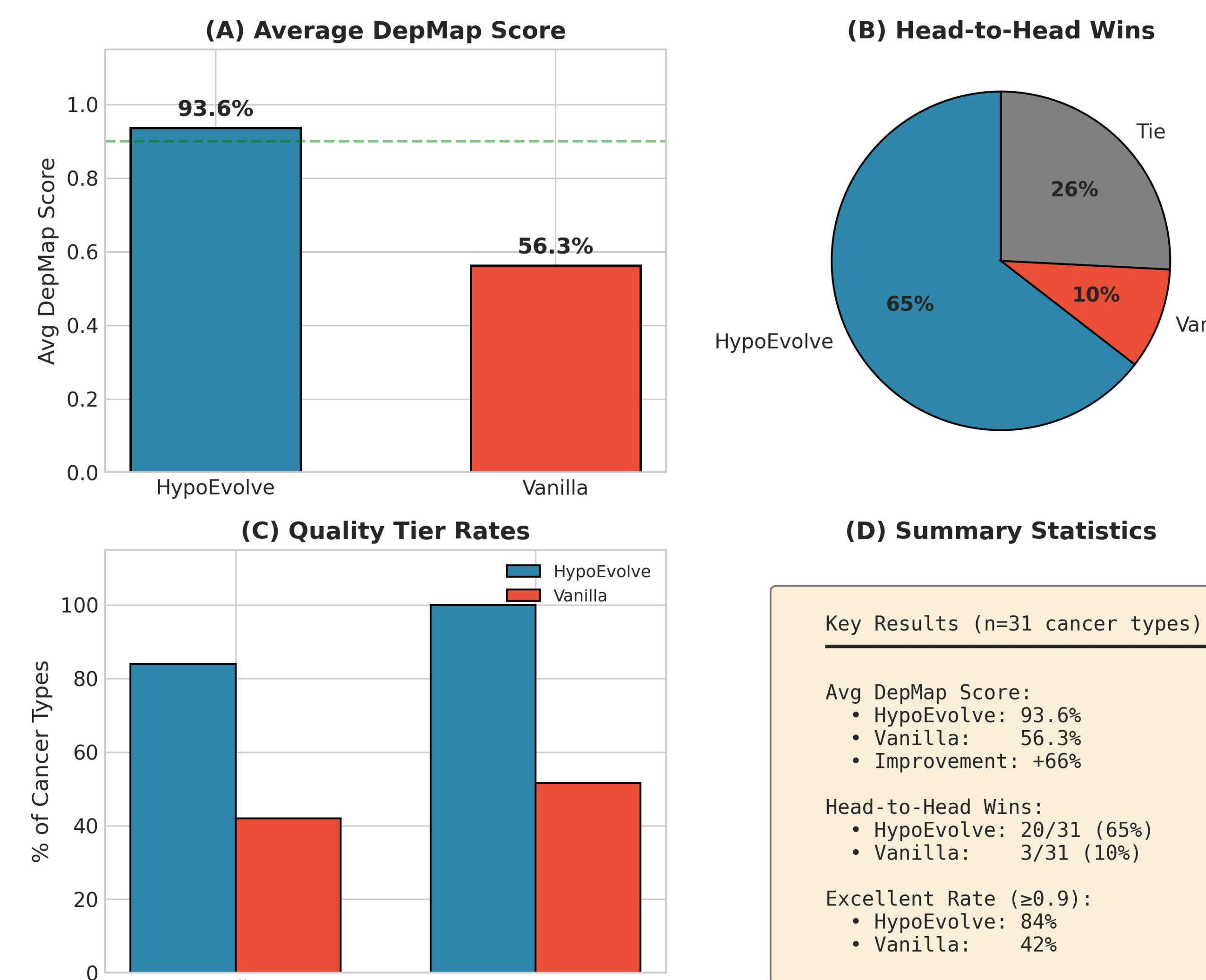


Figure 4: External validation against DepMap CRISPR data across 31 cancer types.

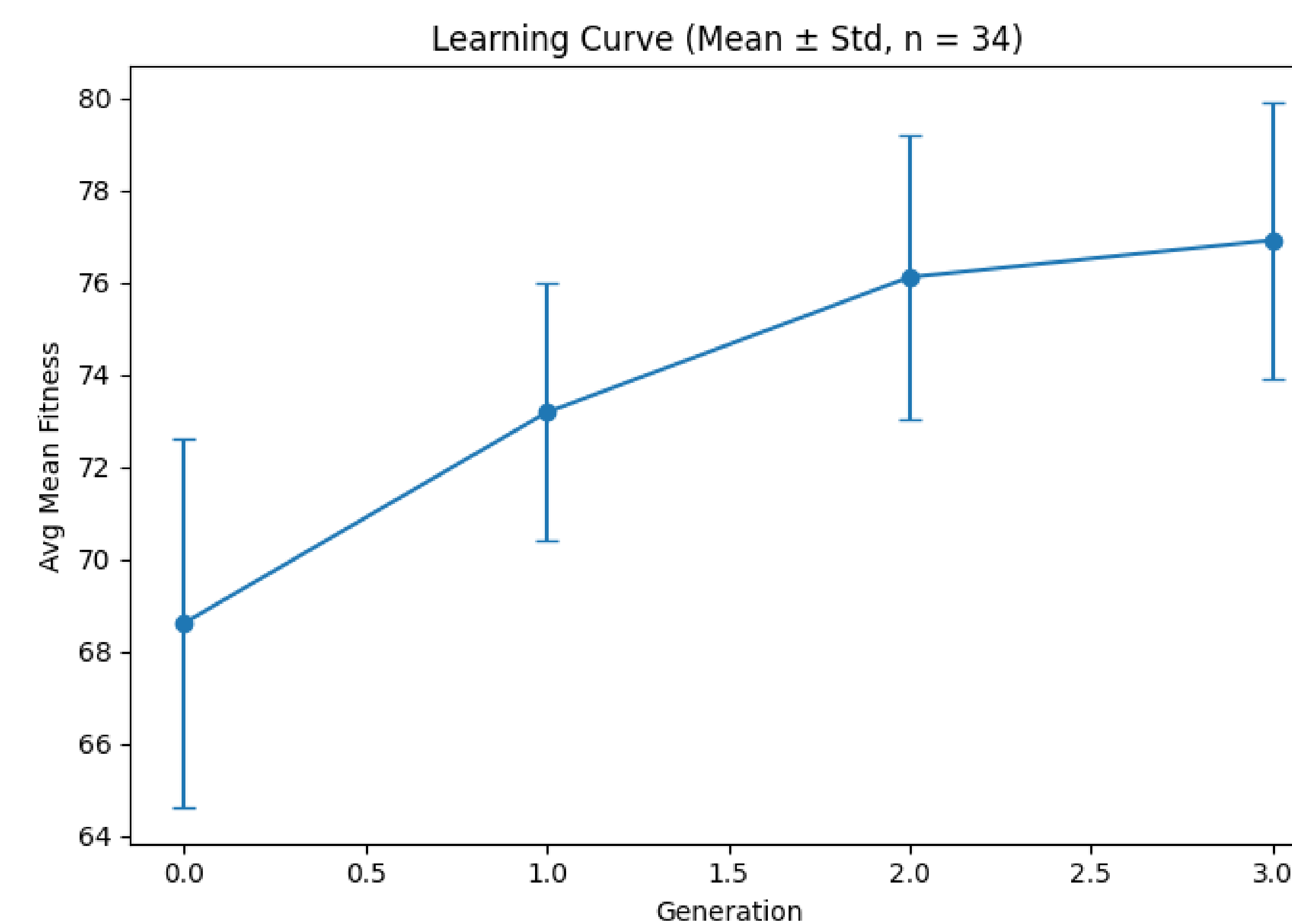


Figure 5: Learning curve showing LLM-evaluated fitness across generations (drug repurposing, 34 cancer types).

Results (T2D)

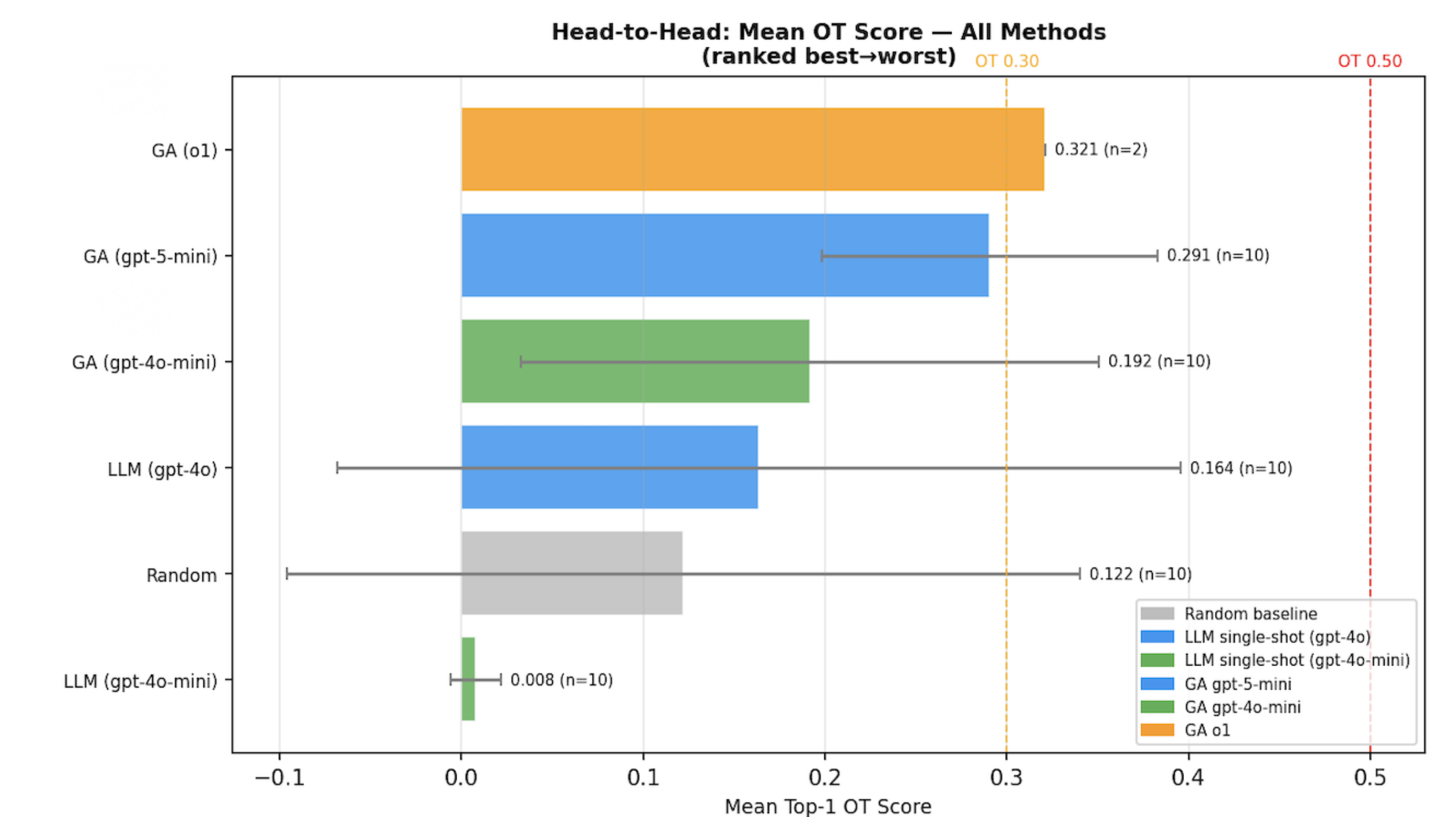


Figure 6: Comparison of single-pass and HypoEvolve with different LLMs.

Key Takeaways

- Evolutionary refinement yields large gains under **external** biological validation.
- Gains are achieved **without DepMap leakage**
- Improvements appear across many cancer types (robustness).
- Framework is general: can be used to target other discovery tasks.

Limitations / Next

- Dependence on LLM-based internal scoring; need stronger calibration / human review.
- Add ablations: remove crossover, remove mutation, remove elitism.
- Include more biological research tasks.

References

- [1] Annu Lambora, Kunal Gupta, and Kriti Chopra. Genetic algorithm-a literature review. In *2019 international conference on machine learning, big data, cloud and parallel computing (COMITCon)*, pages 380–384. IEEE, 2019.